Algorithm for Applying Transformation

Real world coordinate

Convert to homogenous

Perform Transformation

Go back to original coordinate system

In previous lab, transformation was implemented using 'forward mapping'

$$I'(x', y') = I(x, y)$$

or

$$\underbrace{I'(sx, sy) = I(x, y)}$$

Mathematically correct but pratically gives inconsistent results.

Reason:

1) sx, sy were obtained from x, y. It can happen that sx, sy do not lie in image boundary

2) Image generated using

$$\begin{bmatrix} x \\ y \end{bmatrix} = S \begin{bmatrix} x' \\ y' \end{bmatrix}$$

$$S^{-1} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x' \\ y' \end{bmatrix}$$

'forward mapping' gives a grid like pattern in original image

Solution: For all transformations, always use inverse mapping.

We assume (x, y) to be coordinates in new image

↓

The old coordinates are obtained as $S^{-1} \begin{bmatrix} x \\ y \end{bmatrix}$

i.e $$I'(x, y) = I(S^{-1}x, S^{-1}y)$$

This ensures that any pixel in new image is a valid pixel.

↳ It's value is obtained from inverse mapping in old image

Intrinsic Matrix

Rotation, translation and scaling are used to generate extrinsic matrix

(Transformation from real world coordinate to camera coordinate system)

i) The focal length of lens also effect the image formation;

2) The distance of the object also effect the image formation

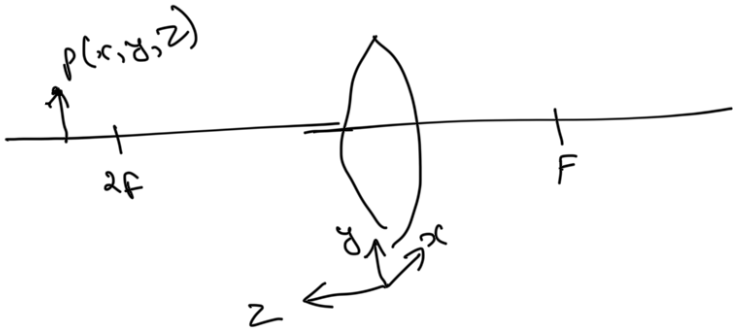
These parameters are captured in an intrinsic matrix

The effect can be understood using ray optics

Consider a convex lens (most CV systems have convex lens

as it is desired that rays should be converging & so that sensor can be placed at focus)

Assume: $Z > 2f$

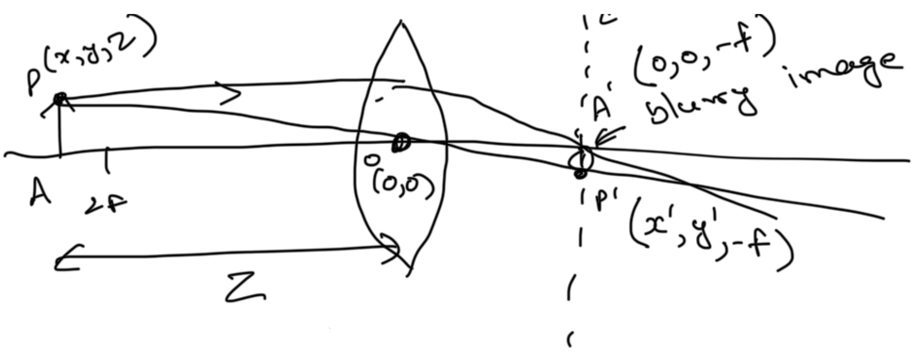


Point P is represented in Camera Coordinate system as it is assumed that P is obtained after extrinsic transformation on the real world object on P'.

Image formation 

// Image Plane

Z: distance of object from camera



observation

$\angle POA = \angle P'OA'$ (vertically opposite)

$\angle PAO = \angle P'A'O$ (90°)

$\Delta POA \sim \Delta P'OA'$ (Angle - Angle test)

$$\frac{PO}{P'O} = \underbrace{\frac{OA}{OA'} = \frac{PA}{P'A'}}$$

$$\frac{Z}{f} = \frac{y}{y'}$$

$$\therefore\ y' = y \times \frac{f}{Z}$$

Similarly

$$\frac{Z}{f} = \frac{x}{x'}$$

$$x' = x \times \frac{f}{Z}$$

i.e Image formation by lens leads to Scaling by $1/z$ in Coordinates

In terms of matrix Transformation

$$\begin{bmatrix} x' \end{bmatrix} \quad \begin{bmatrix} \frac{f}{z} & 0 \end{bmatrix} \begin{bmatrix} x \end{bmatrix}$$

$$\begin{bmatrix} \\ y' \end{bmatrix} = \begin{bmatrix} & \\ 0 & \frac{f}{z} \end{bmatrix} \begin{bmatrix} \\ y \end{bmatrix}$$

Using homogenous coordinate

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} f/z & 0 & 0 \\ 0 & f/z & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x/z \\ y/z \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \frac{1}{z} \underbrace{\begin{bmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\text{Intrinsic matrix}} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

x', y' on RHS are in physical units (mm)

↓

for image x', y' are in pixels

Let the sensor grid be as follows

P_x : distance between two pixels in mm in x - direction

P_y : distance between two pixels in mm in y - direction

u P_x

x'

$u = \frac{x'}{P_x}$ $v = \frac{y'}{P_y}$

— px py

(u, v) ⇒ coordinates in image plane (in terms of pixels)

$$\underbrace{\begin{bmatrix} u \\ v \\ 1 \end{bmatrix}}_{\substack{\text{2D} \\ \text{image} \\ \text{coordinate}}} = \frac{1}{Z} \underbrace{\begin{bmatrix} \frac{f}{p_x} & 0 & 0 \\ 0 & \frac{f}{p_y} & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\underline{\text{Intrinsic Matrix}}} \underbrace{\begin{bmatrix} x \\ y \\ Z \end{bmatrix}}_{\substack{\text{3D} \\ \text{Camera} \\ \text{Coordinate}}}$$

Also known as
Perspective transformation
|
observe $1/Z$ term ('heart' of 3D graphics)
|
Magnification for near objects is high
/
far away object appear to be <u>smaller</u>.